

FC_IDE User Guide

Rev.2.0

Revision History

Revision	Author	Date	Changes
1.0		2022/04/26	Initial release
1.1.0		2022/11/15	- Updated the description of <i>Section 3.4.1 Build Project</i>; - Updated the note in Step 5 of <i>Section 3.5.2 Debug</i>; - Added <i>Appendix A</i>, and moved the note about how to load the register definition file into FC_IDE from <i>Section 3.5.2 Debug</i> to <i>Appendix A</i>; - Fixed the typo "Short View" and changed it to "Show View" in <i>Appendix A</i>; - Editorial changes.
1.1.1		2022/11/29	Updated the note in Step 5 of <i>Section 3.5.2 Debug</i>
1.1.2		2022/12/27	Change the name of "Easy IDE" to "FC_IDE"
1.2		2022/12/30	- Added information for FC7300; - Updated the Build Project description for FC7300 in <i>Section 3.4.1 Build Project</i>; - Updated the note for FC7300 in Step 5 of <i>Section 3.5.2 Debug</i>.
2.0		2023/02/10	- Added a new section <i>New Project</i> for creating new projects; - Updated the <i>Note: How to deal with warnings after importing an existing project</i> in <i>Section 3.3 Import Project</i>; - Updated the debug steps in <i>Section 3.5.2 Debug</i> - Added a new method for creating debugging configuration; - Updated the <i>Note: How to deal with debug problems</i>

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Chapter 1 Introduction

This user guide provides basic information on how to start up the FC_IDE, and how to import, build, clean, debug and edit projects in the IDE.

Chapter 2 Launch

Before running the FC_IDE on your computer, take the following steps:

- 1) Extract the **FC_IDE** ZIP file to the directory where you want to place the IDE.
- 2) After extraction, double-click the EXE file to run it.

Note:

- After extraction, the IDE can be run without installation.
- For quick and easy access next time, you can right-click the EXE file, and select **Pin to Start** or **Pin to taskbar**.

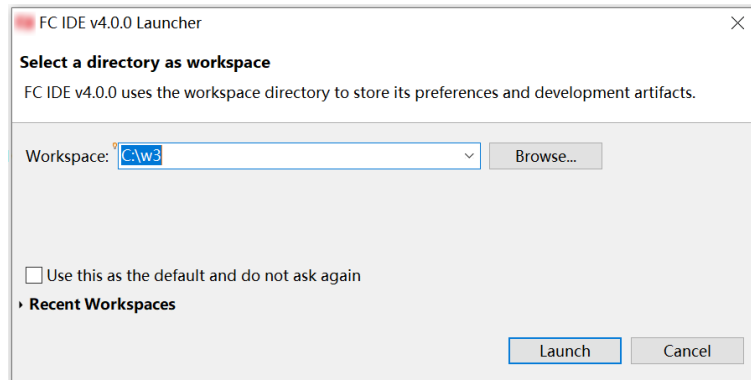
AStyle	2022/11/21 13:29	文件夹	
cmd	2022/11/21 13:29	文件夹	
configuration	2022/12/29 9:44	文件夹	
Devices	2022/11/21 13:30	文件夹	
dropins	2021/6/12 20:16	文件夹	
features	2022/11/21 13:30	文件夹	
gcc-10.2.1_20201103	2022/11/21 13:30	文件夹	
Graphviz	2022/11/21 13:30	文件夹	
JLink	2022/12/28 17:12	文件夹	
p2	2022/11/21 13:30	文件夹	
plugins	2022/12/28 17:14	文件夹	
readme	2022/11/21 13:30	文件夹	
.eclipseproduct	2021/6/11 20:06	ECLIPSEPRODUCT ...	1 KB
artifacts.xml	2022/12/29 9:44	XML 源文件	141 KB
eclipse.exe	2021/6/12 20:18	应用程序	129 KB
Flagchip_FC_IDE.exe	2021/6/30 10:35	应用程序	408 KB
Flagchip_FC_IDE.ini	2022/11/21 11:27	配置设置	1 KB
notice.html	2021/6/8 1:20	Microsoft Edge HT...	10 KB
ReleaseNote.txt	2022/12/27 14:25	文本文档	1 KB

Chapter 3 Basic Functions

This chapter describes the basic functions of the IDE that you may frequently use.

3.1 Launch IDE for the First Time

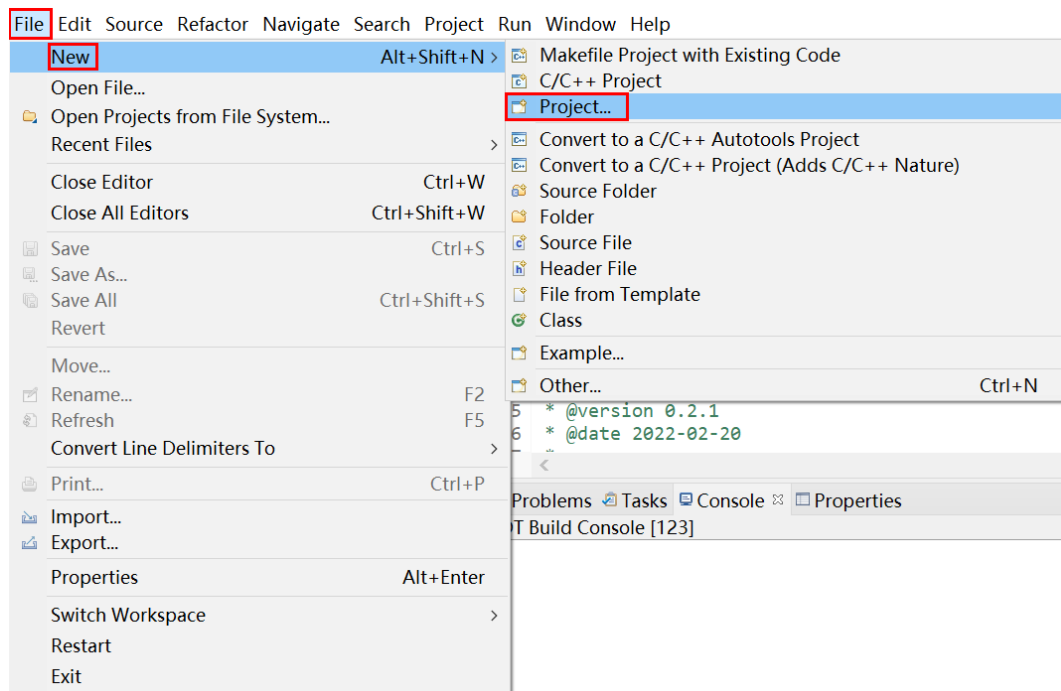
The first time you start up the IDE, you are prompted to specify a location to store your workspace. By default, the IDE puts your workspace in `C:/Projects/test`. If you want to store your workspace in another location, click **Browse** and select a directory. Then click **Launch** to start up the software.



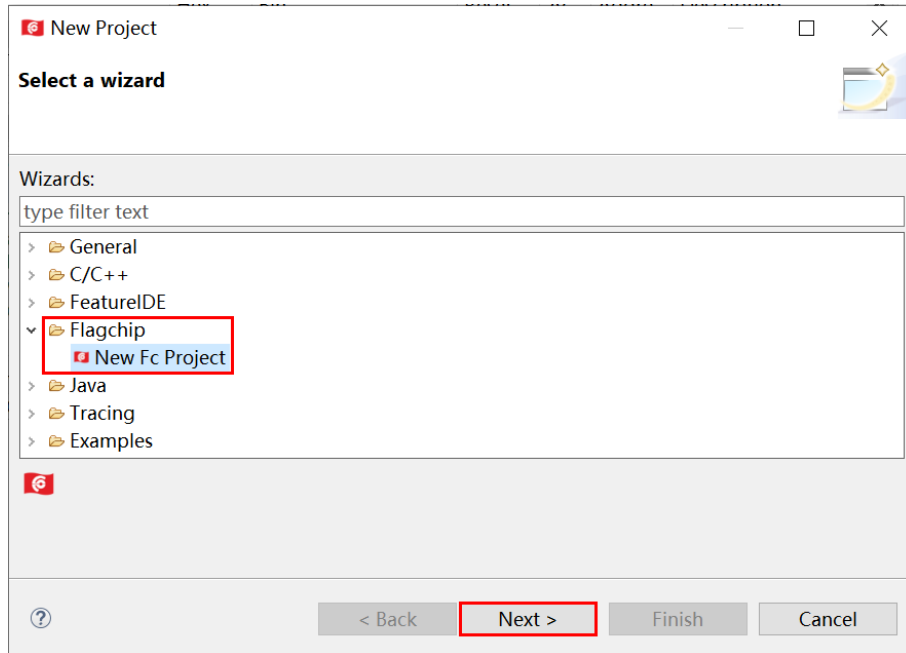
3.2 New Project

To create a new project, take the following steps:

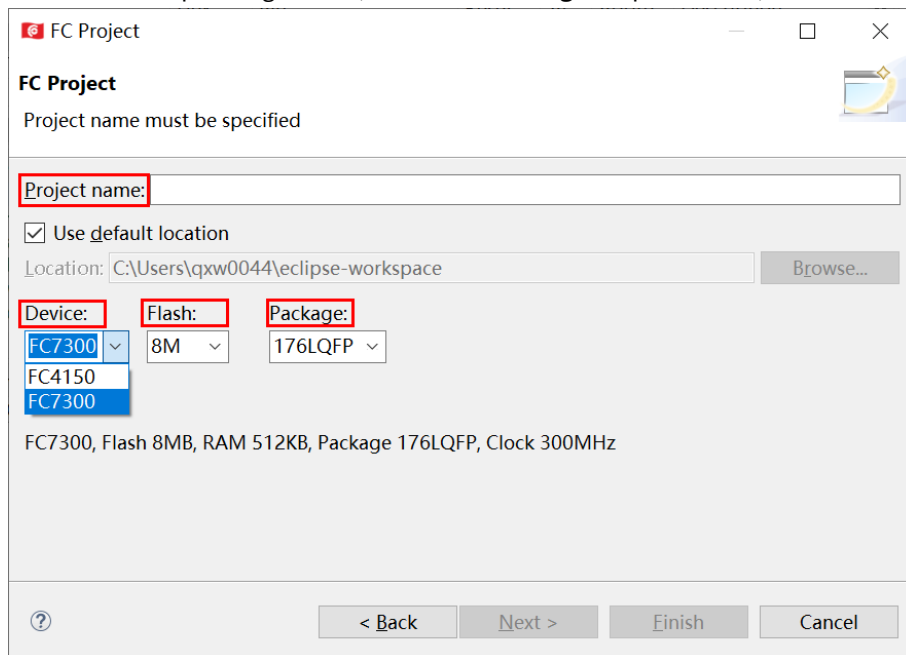
- 1) Select **File > New > Project**.



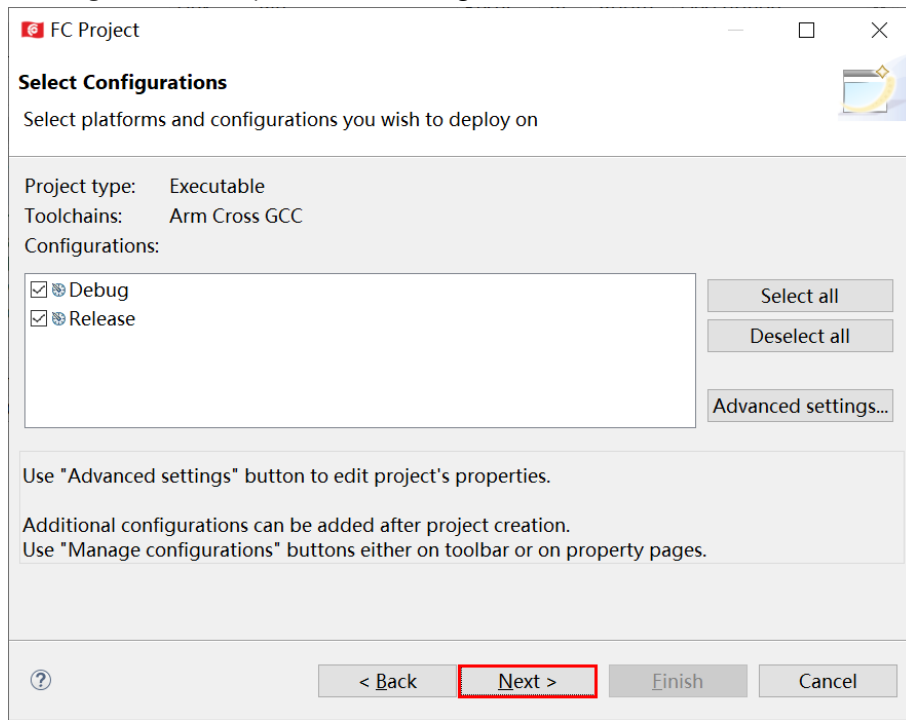
- 2) In the pop-up **New Project** dialog, select **Flagchip > New Fc project**, and click **Next**.



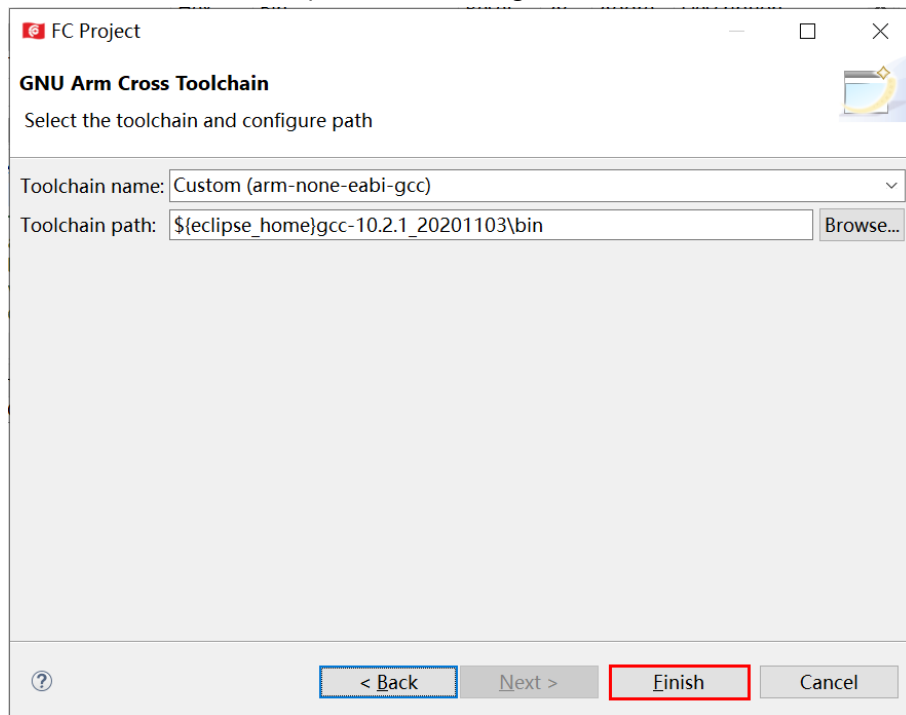
- 3) In the pop-up **FC Project** dialog, enter your project name into the **Project name** text box, select the device specification from the corresponding **Device**, **Flash** and **Package** drop-down list, and click **Next**.



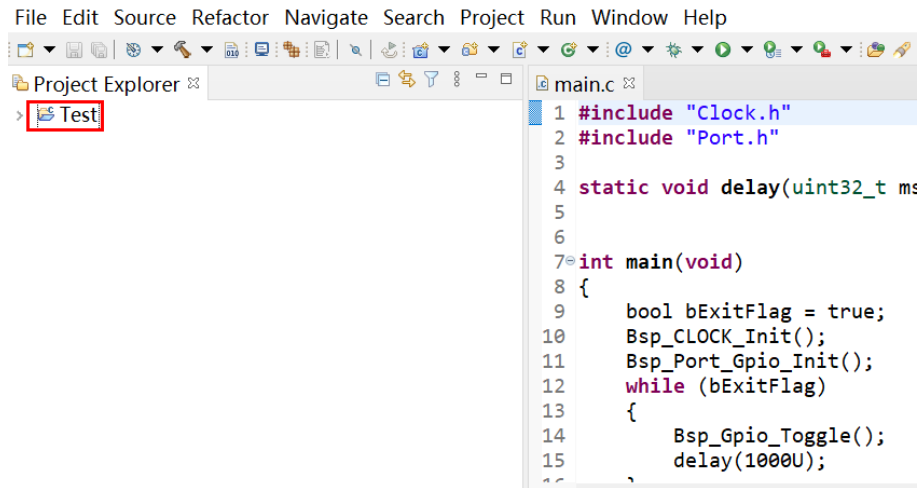
- 4) For the **Select Configurations**, keep the default configuration, and click **Next**.



- 5) For the **GNU Arm Cross Tool Chain**, keep the default configuration, and click **Finish**.



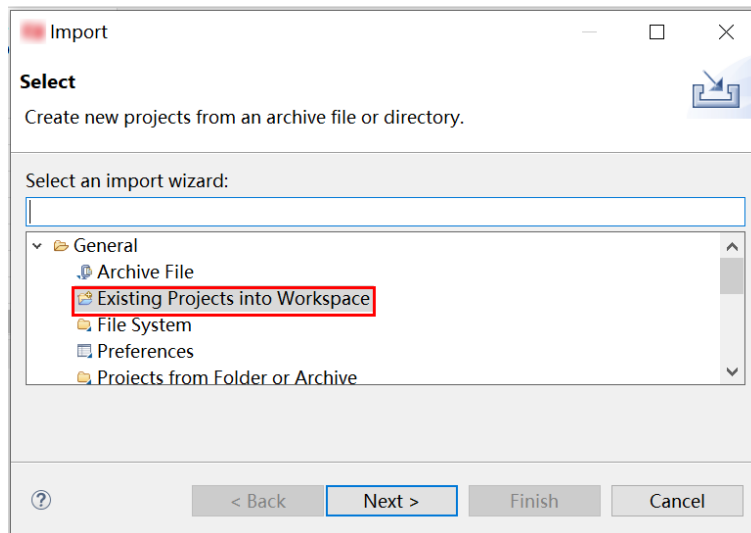
- 6) The project you created will be shown under the **Project Explorer** in the left pane.



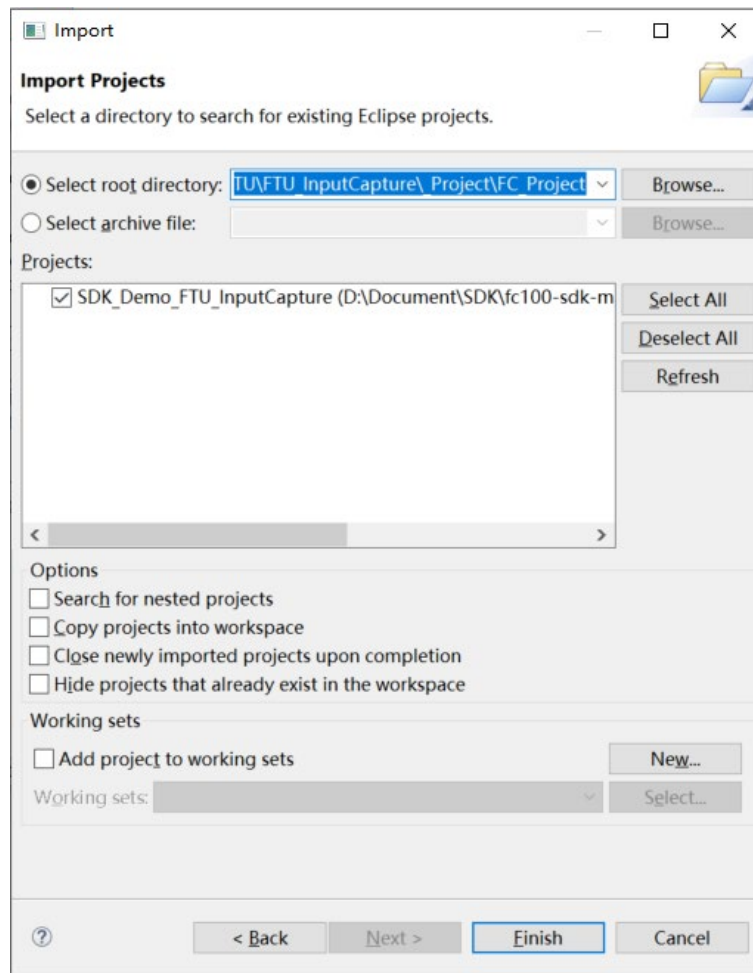
3.3 Import Project

To view the code of an existing project, you can import the project to the IDE by taking the following steps:

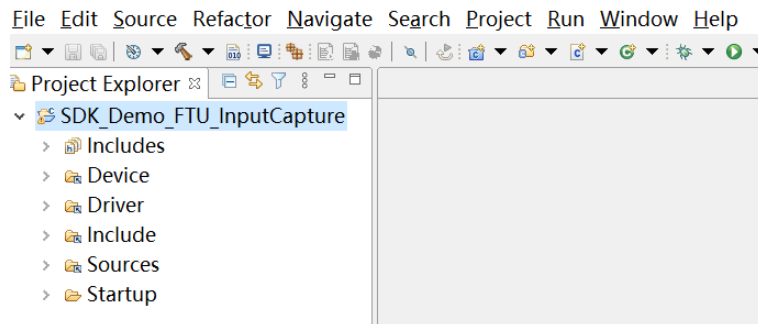
- 1) Select **File > Import**.
- 2) In the **Import** dialog, select **General > Existing Projects into Workspace**, and click **Next**.



- 3) Enter the project path into the **Select root directory** text box, or click **Browse** to select a directory. The imported project will be displayed under **Projects**.



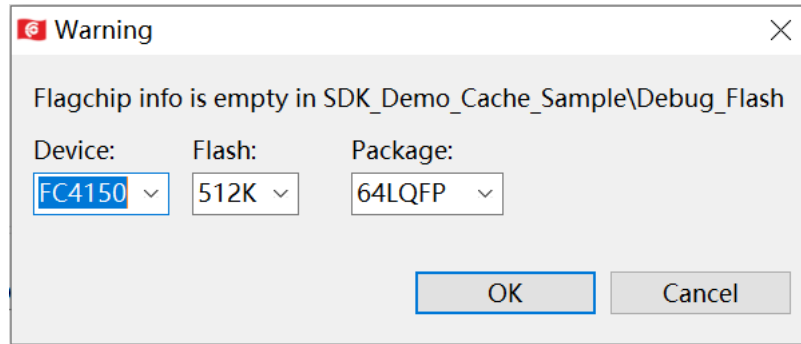
- 4) Click **Finish**, and the project will be shown on the left pane.



Note: How to deal with warnings after importing an existing project.

For the new version of FC_IDE, if there is no device information configured for a project you imported, a **Warning** dialog will pop up after you importing the project.

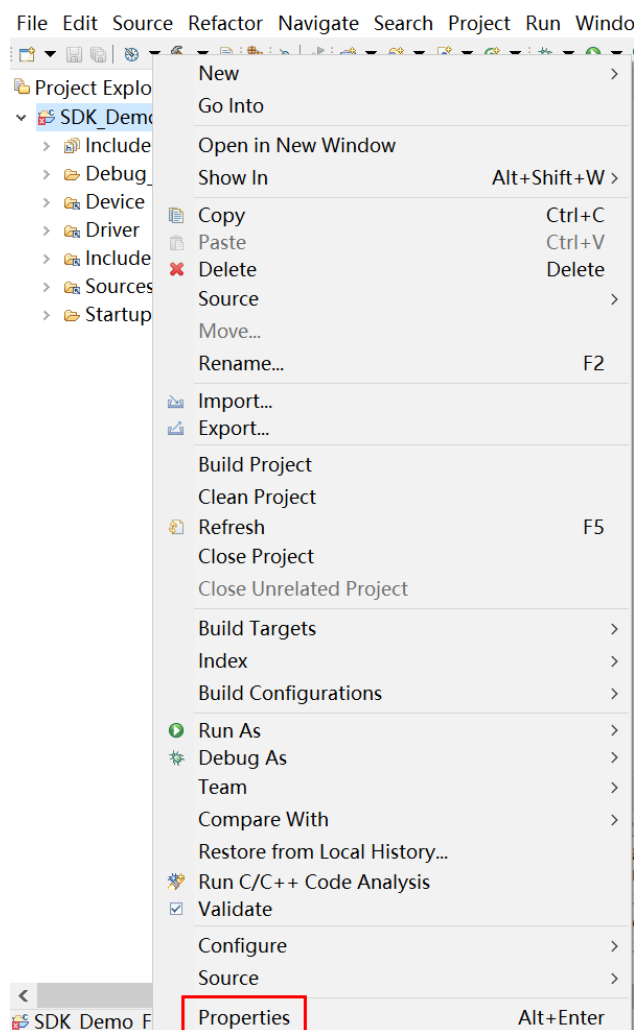
In this case, select the device specification from the corresponding **Device**, **Flash** and **Package** drop-down list, and click **OK**. Then, you can build the project as usual.



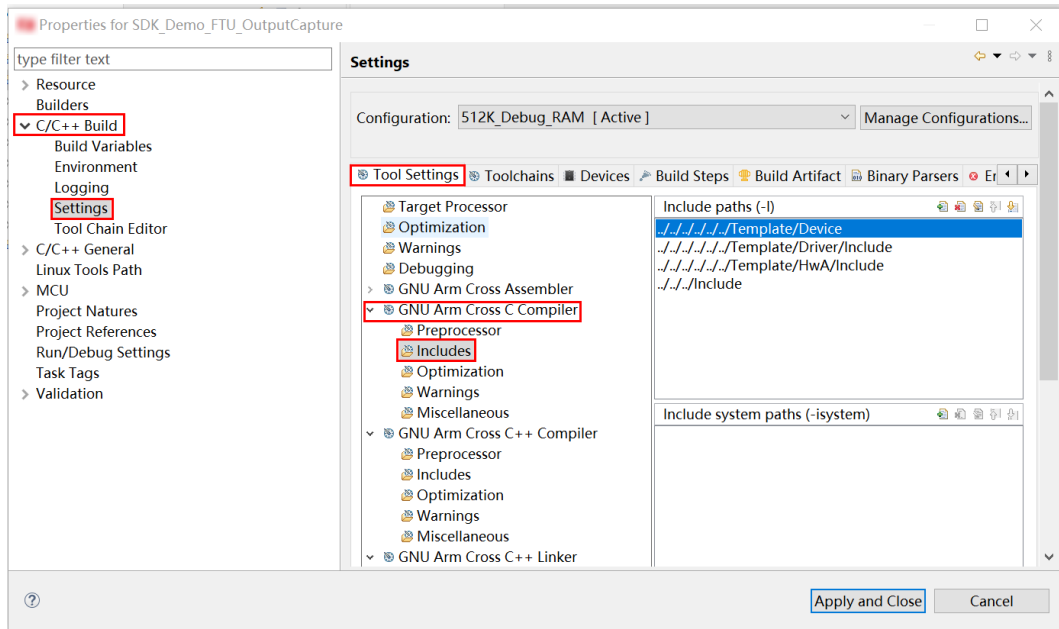
3.4 Add Header Files to Includes

For quick reference of header files you may need, you can add them to the Includes folder of the project by taking the following steps:

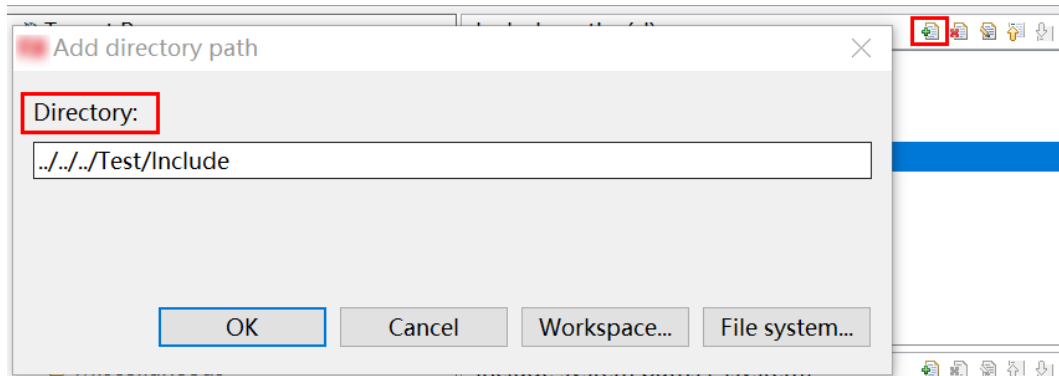
- 1) Right-click the project name, and select **Properties** from the pop-up menu.



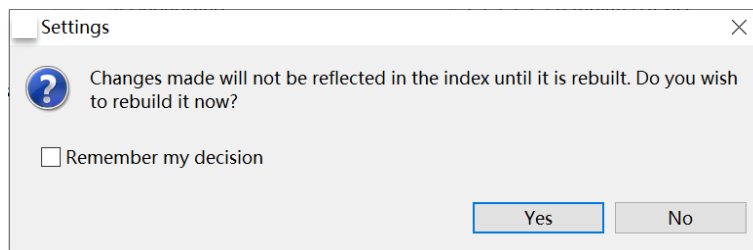
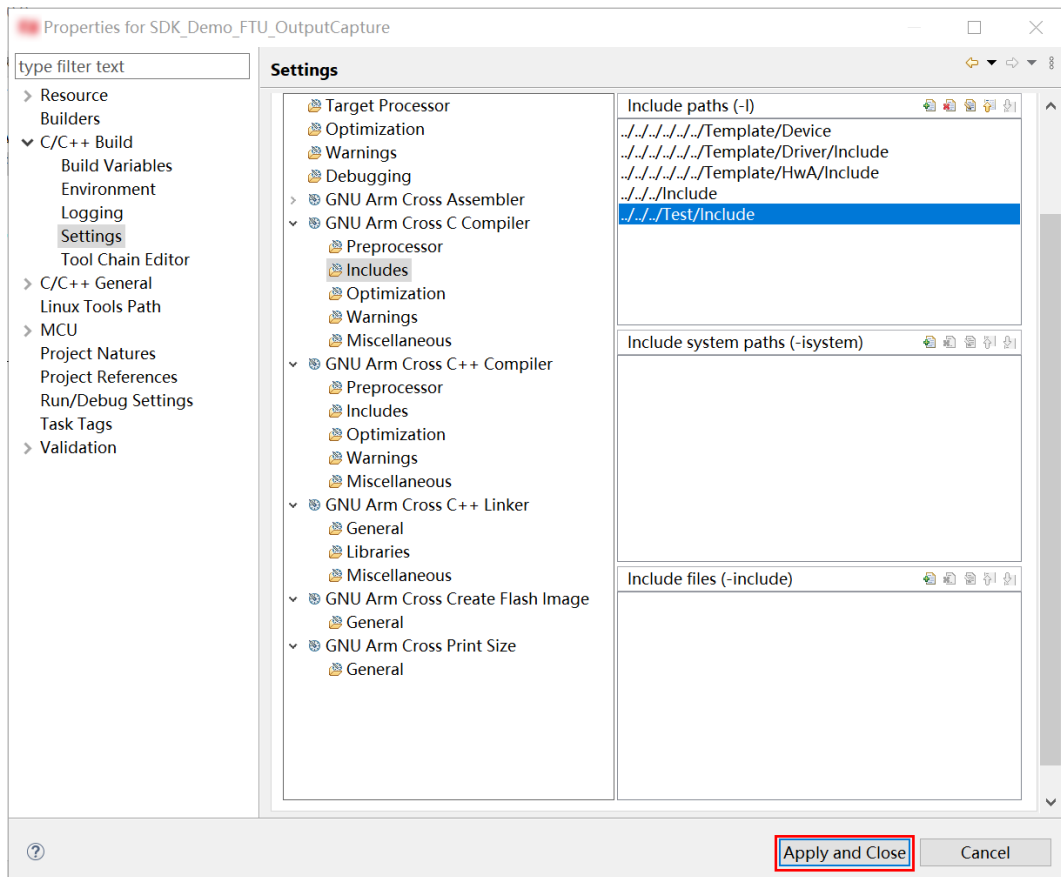
- 2) In the pop-up dialog, select **C/C++ Build > Settings** on the left pane.
- 3) In the **Tool Settings** tab on the right pane, select **GNU Arm Cross C Compiler > Includes**.



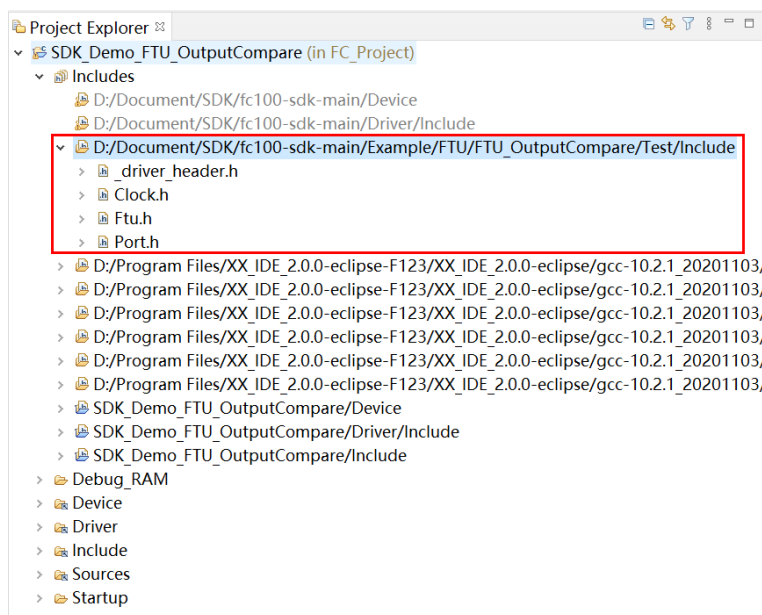
- 4) Click the Add icon () on the top right corner of the tab, and in the **Add directory path** dialog, enter your header file path into the **Directory** text box and click **OK**.



- 5) Click **Apply and Close**, and in the **Settings** dialog, click **YES** to rebuild the project immediately.



6) After rebuilding, the header files will be displayed under the Includes folder.



3.5 Build/Clean Project

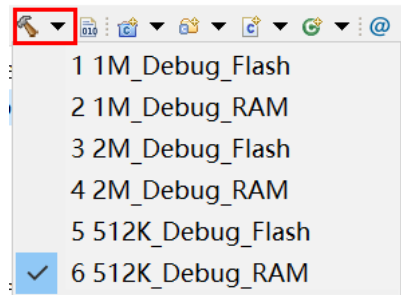
3.4.1 Build Project

There are two major kinds of build configuration options: **Debug_Flash** and **Debug_RAM**. The major difference between the two options is the location where the program is loaded. If **Debug_Flash** is selected, the data will be loaded to the flash; if **Debug_RAM** is selected, the data will be loaded to the RAM.

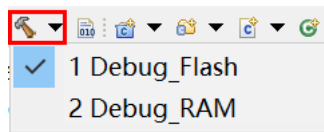
Note: For FC4150, we provide another three choices 512K, 1M and 2M for each option: 512K Debug_Flash and 512K Debug_RAM are for FC4150F512K; 1M Debug_Flash and 1M Debug_RAM for FC4150F1M; and 2M Debug_Flash and 2M Debug_RAM for FC4150F2M.

Before starting to build, you can select one of the following build configurations for your project by clicking the upside-down triangle icon (▼) next to the icon (🔧) in the toolbar.

- For FC4150, the configurations are as follows:

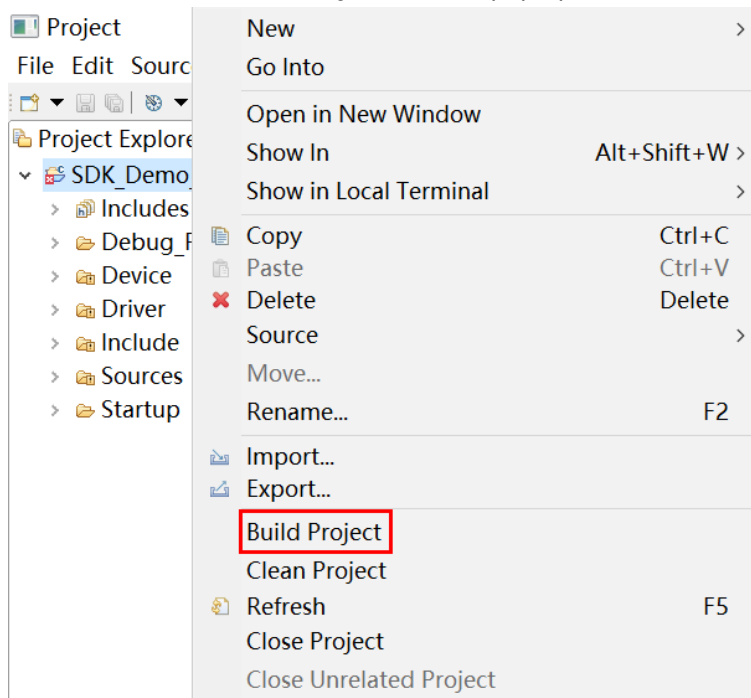


- For FC7300, the configurations are as follows:

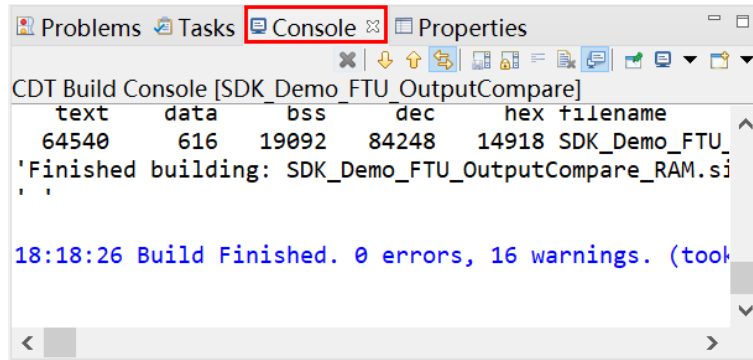


Taking "Debug_RAM" as an example, the build steps are as follows:

- 1) Right-click the project name, and select **Build Project** from the pop-up menu.



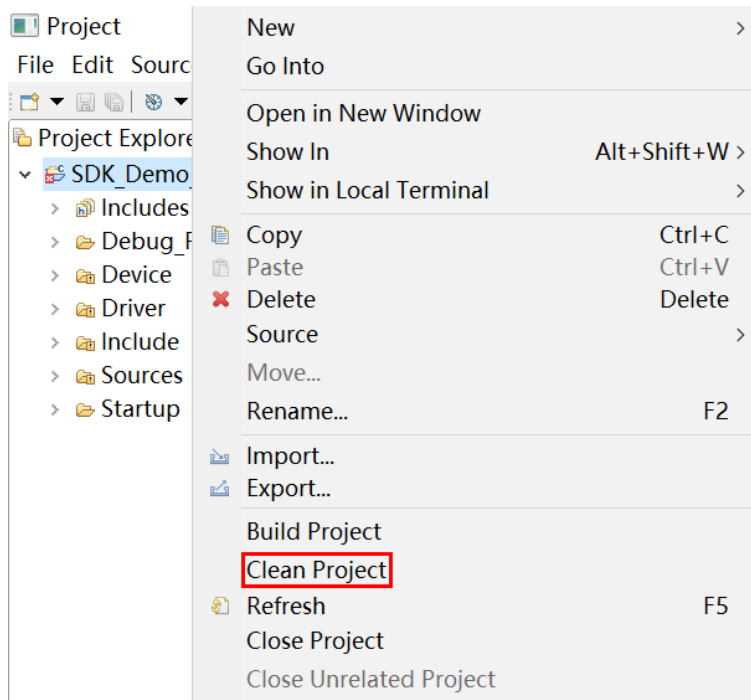
- 2) The build result will be displayed on the **Console** tab at the lower part of the window.



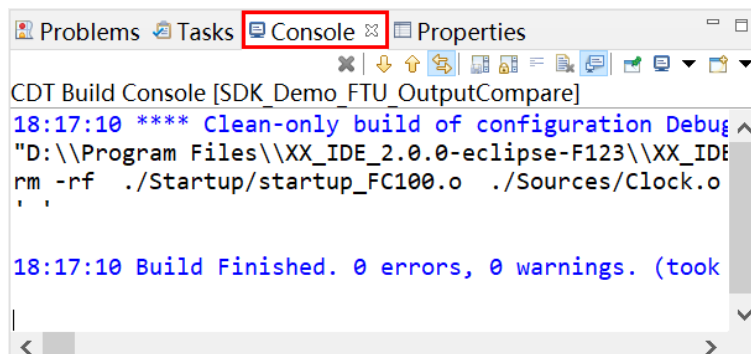
3.4.2 Clean Project

To clean the project, take the following steps:

- 1) Right-click the project name, and select **Clean Project** from the pop-up menu.



- 2) The clean result will be displayed on the **Console** tab at the lower part of the window.



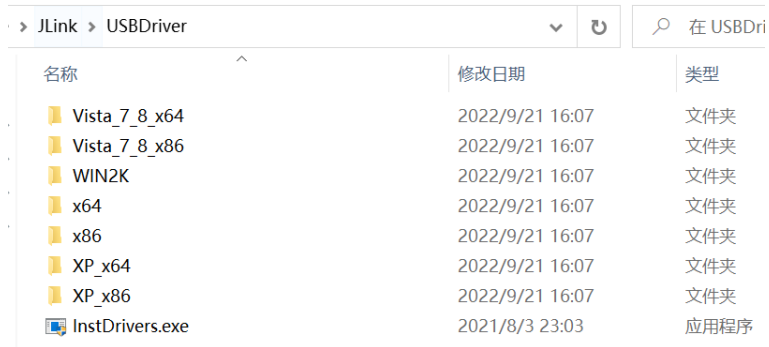
3.6 Debug Project

Only JLink debug is supported currently. If you haven't used the JLink before, you must install the JLink USB driver before debugging. If you already have the driver installed, you can jump to Chapter 3.5.2 Debug.

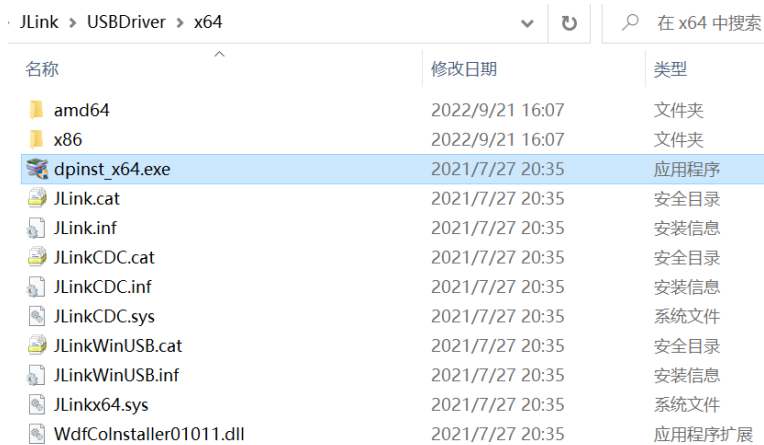
3.5.1 Install JLink USB Driver

To install the USB driver, take the following steps:

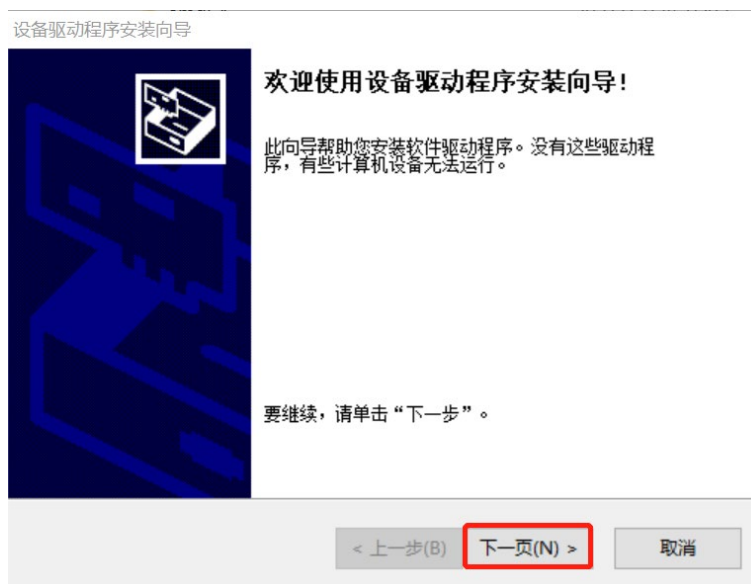
- 1) Open **JLink\USBDriver** in the FC_IDE folder you've extracted, and open a folder corresponding to your computer's operating system.



- 2) Double-click the **dpinst_xxx.exe** file, then the installation wizard will pop up.



- 3) Click **Next** in the installation wizard.



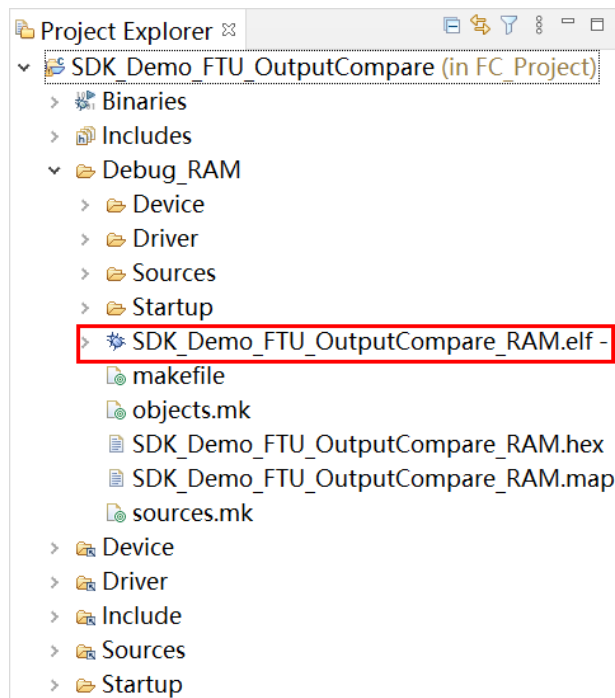
- 4) Click **Finish** to complete the installation.



3.5.2 Debug

For the first time you debug the project, take the following steps:

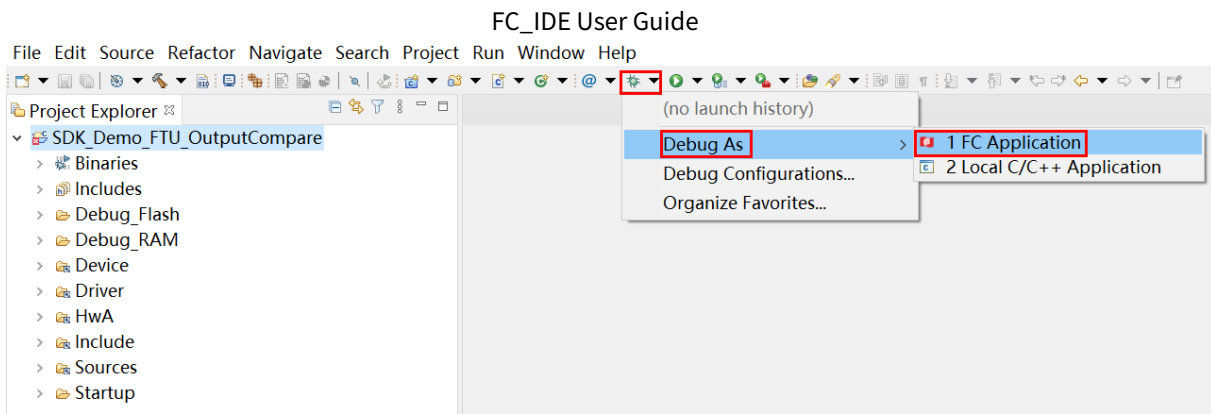
- 1) Right-click the project name and select **Refresh** from the pop-up menu, then an ".elf" file will be shown under the folder Debug_RAM.



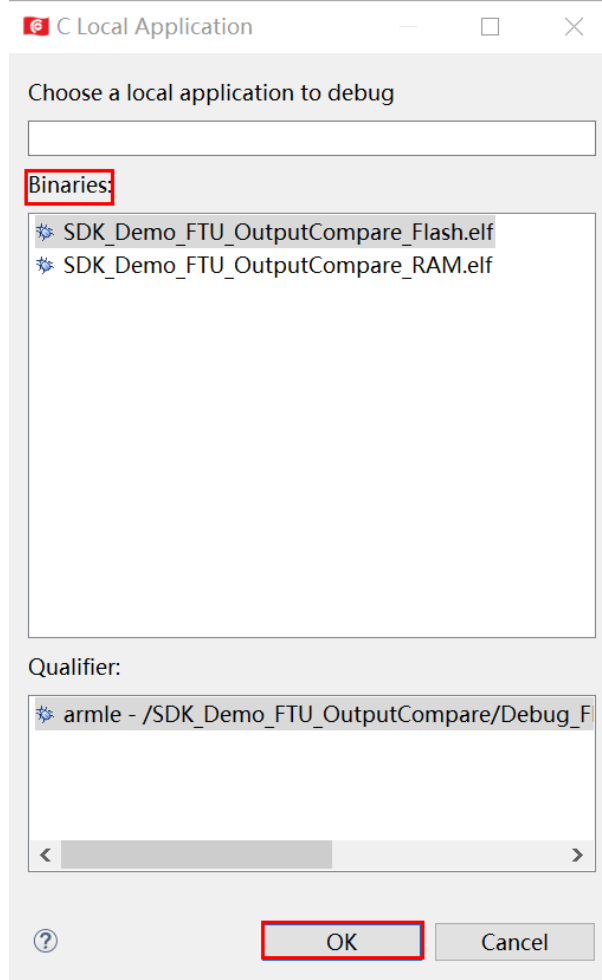
- 2) Create configuration and debug. There are two ways to create configuration and debug for a project.

Method 1:

- a) Click the upside-down triangle icon (▼) next to the Debug icon (🐞) in the toolbar, and select **Debug As > FC Application** from the drop-down list.



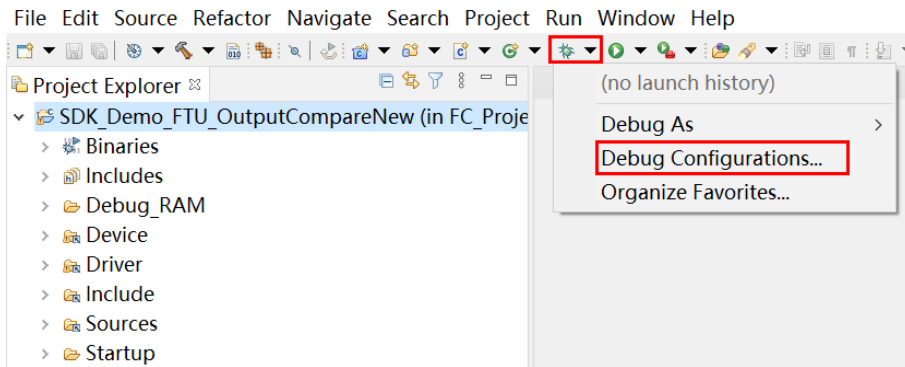
- b) In the pop-up **C Local Application** dialog, double-click one .elf file you need. Then the debug begins.



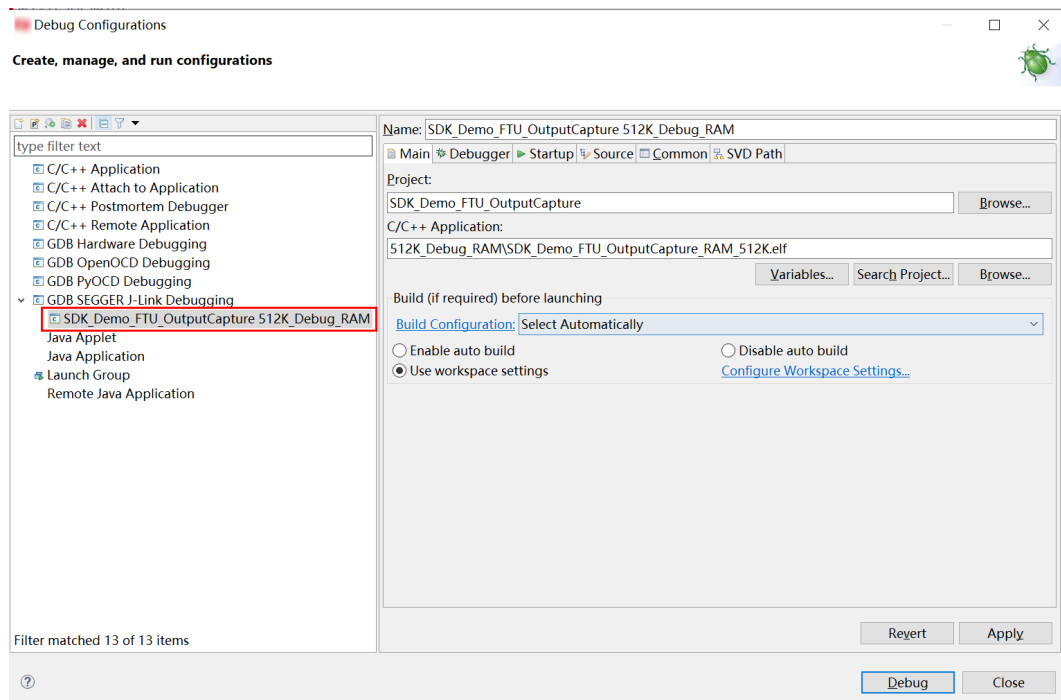
Note: If there is only one debug option for a project, there will be no pop-up dialog, and the debug will start immediately.

Method 2:

- a) Click the upside-down triangle icon (▼) next to the Debug icon (🐞) in the toolbar, and select **Debug Configurations** from the drop-down list.



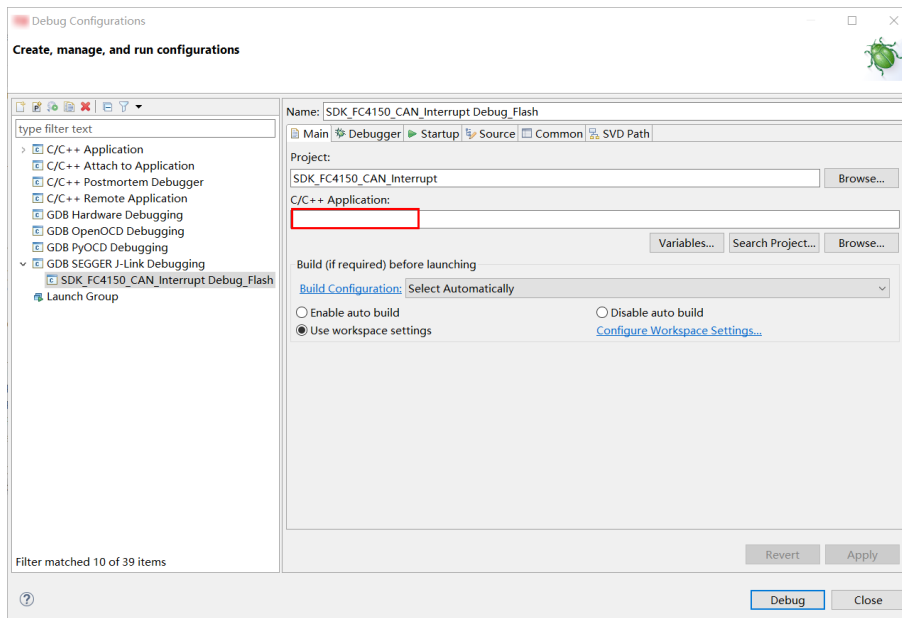
- b) In the **Debug Configurations** dialog, double-click **GDB SEGGER J-Link Debugging** on the left pane to create a new configuration.



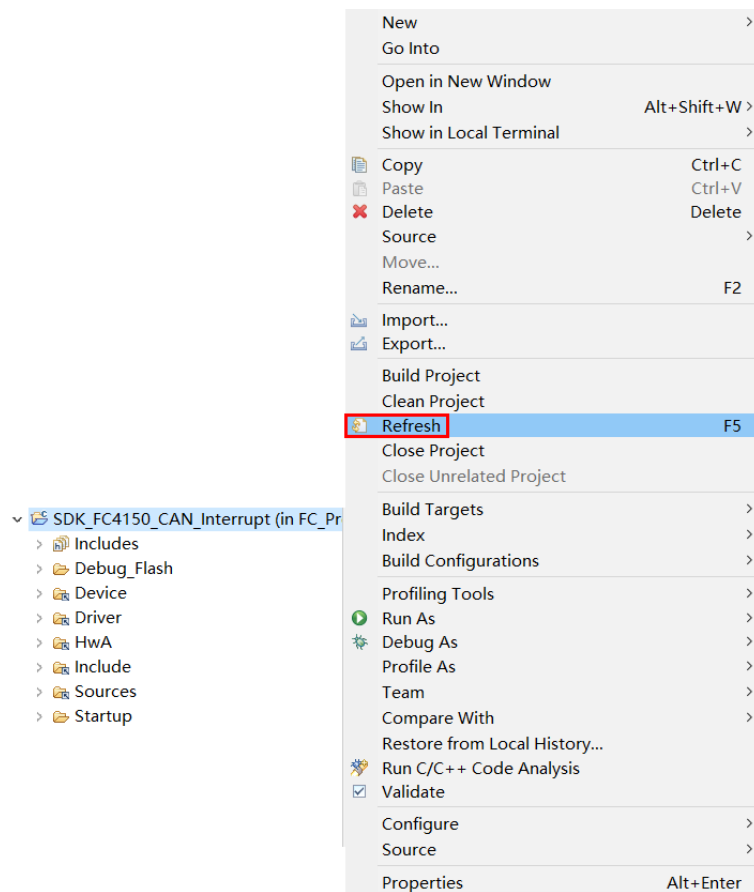
- c) Click **Debug** to start debugging.

Note: What to do if there is no .elf file

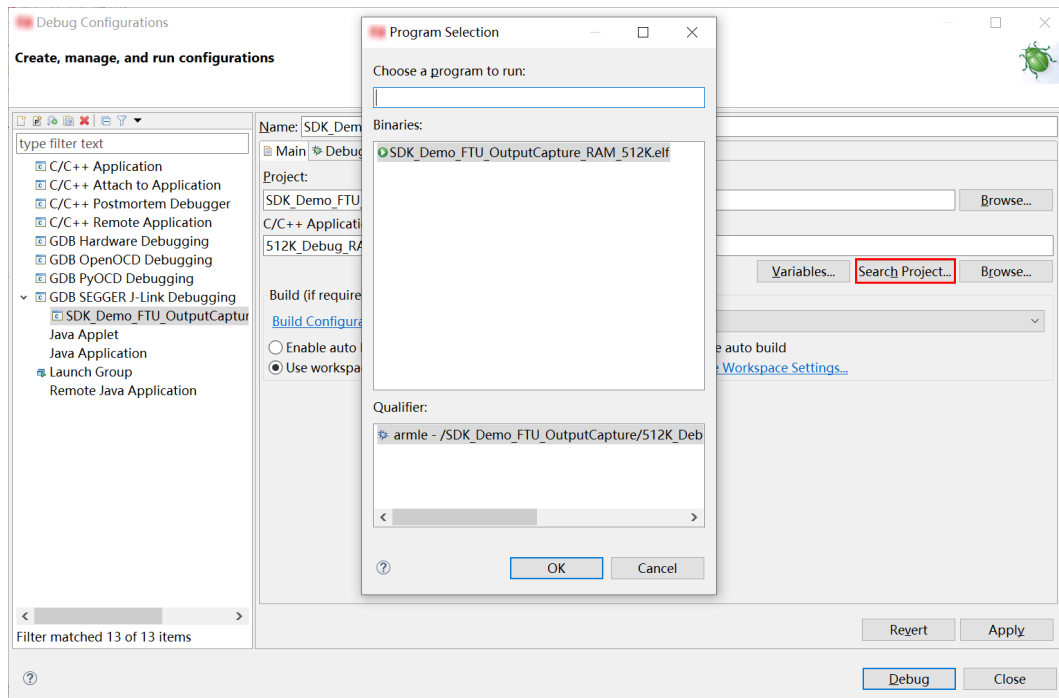
If there is no **.elf** file in the **C/C++ Application** text box when you debug for the first time, take the following steps:



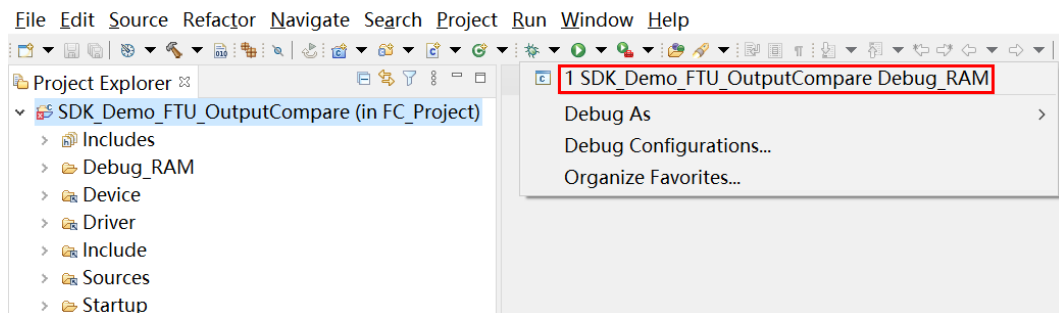
- A. Right-click on the project, and select **Refresh** from the pop-up menu.



- B. Click **Search Project**, select the project from the **Program Selection** dialog, and click **OK**.

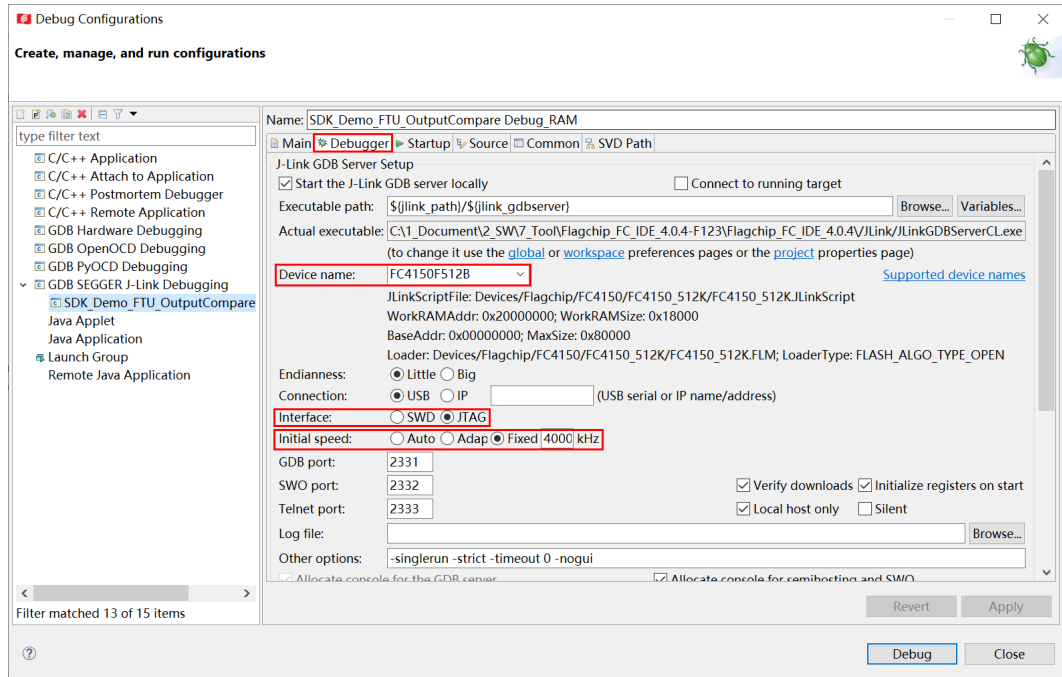


- 3) For the next time you want to debug the project, you only need to select the configuration you saved from the Debug drop-down list, then the debug begins.



Note: How to deal with debugging problems.

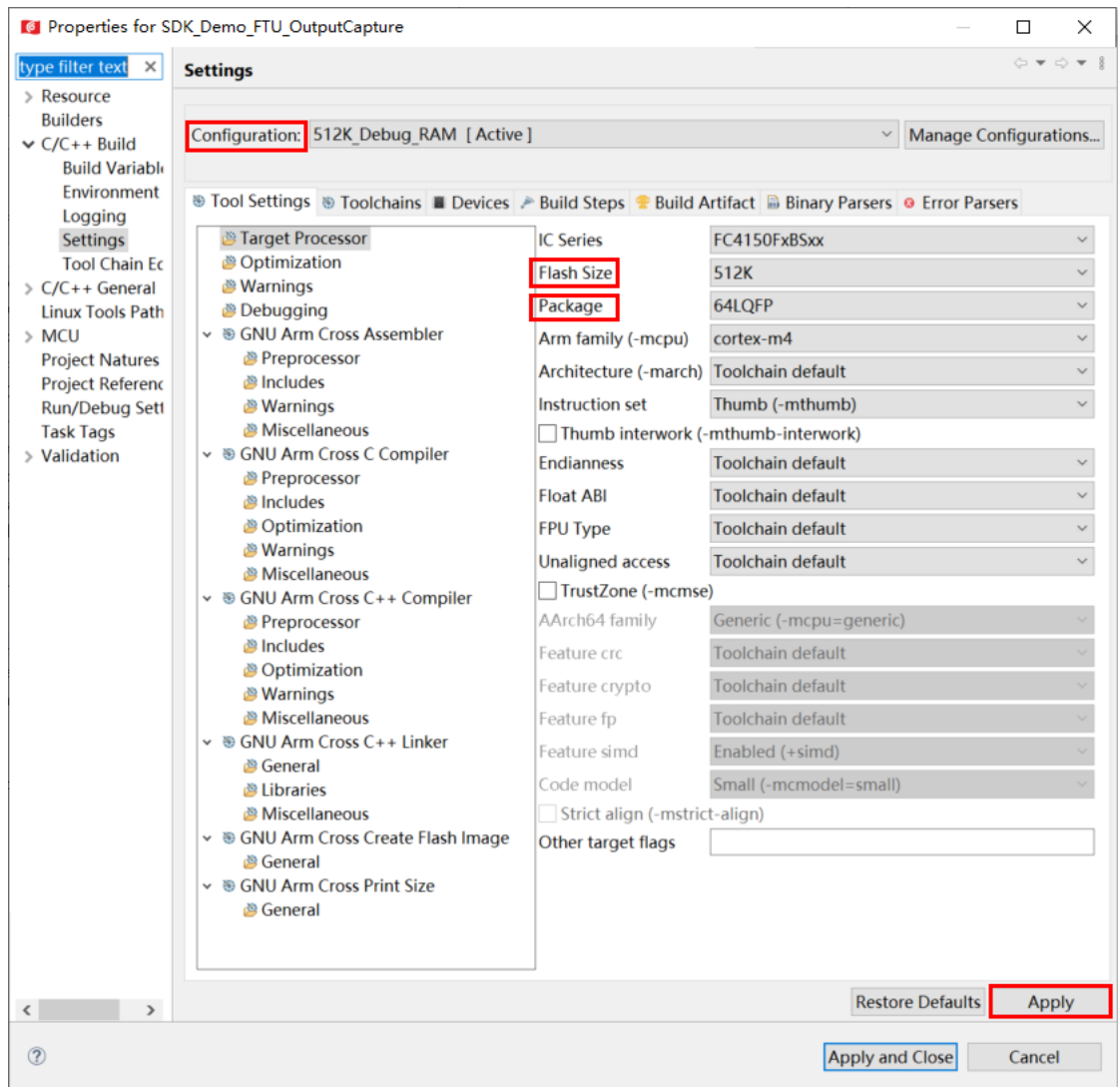
- If you encountered debugging problems after creating a GDB SEGGER J-Link Debugging configuration:
 - a) In the **Debugger** tab, check whether the **Device Name** and **Interface** are correctly configured.



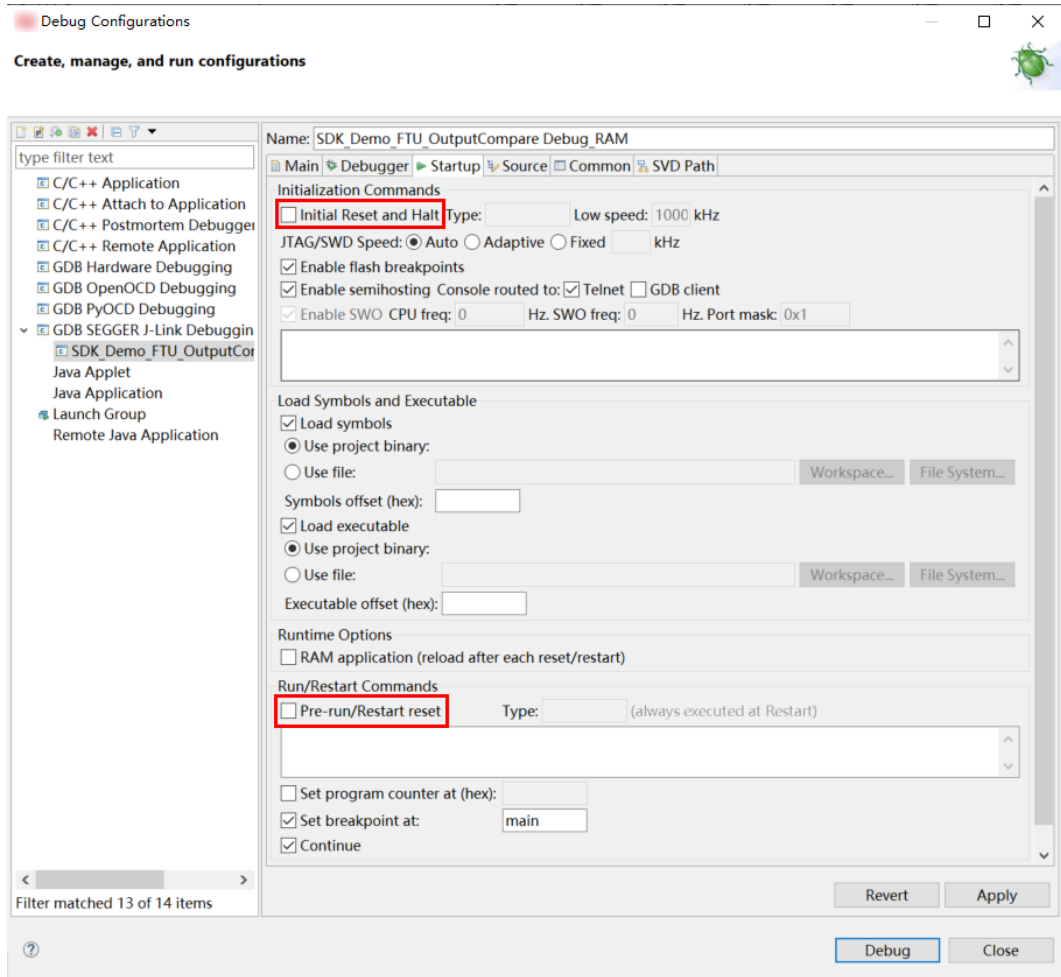
- A. **Device name** (For FC4150: there are FC4150F512B, FC4150F1MB, and **FC4150F2MB**; For FC7300, there are FC7300F8MD_CORE0, FC7300F8MD_CORE1, and FC7300F8MD_CORE2; the device name here must be matched with the build configuration you've selected in *Section 3.4.1 Build Project*).

If there is a mismatch of device name, take the following steps:

- i. Right-click on the project name, select **Properties** from the pop-up menu.
- ii. Select the right device information from the **Configuration**, **Flash Size**, and **Package** drop-down list, and click **Apply**.



- iii. Return to the **Debugger** tab, you can select the device name from the **Device name** drop-down list.
 - B. **Interface** (should be **JTAG**).
 - C. **Initial speed** (Should be set as fixed **4000** kHz).
- b) In the **Startup** tab, ensure both the **Initial Reset and Halt** and **Pre-run/Restart reset** are unchecked. If you make any changes, click **Apply**.



- The register information of FC4150/FC7300 is also provided in XML format for convenient debugging. For details on how to load the file into the FC_IDE, refer to the [Appendix A](#).

3.7 Edit Project

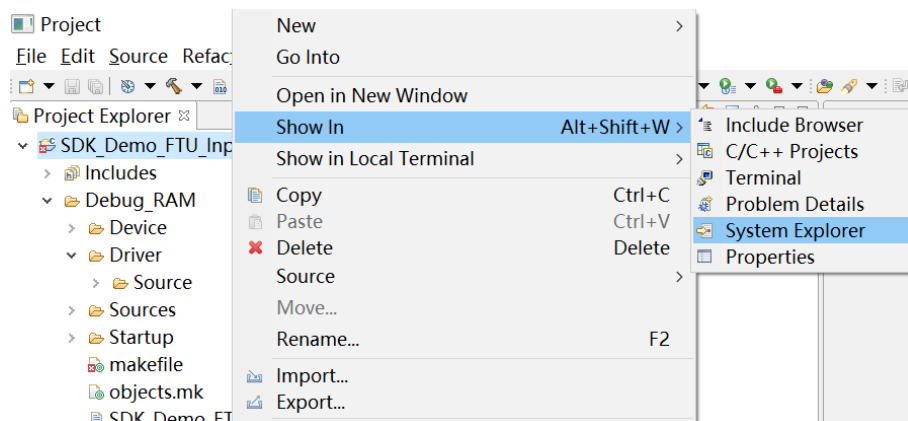
You can edit the project you imported, such as changing project name, creating a new file/folder and adding a file/folder.

3.5.3 Change Project Name

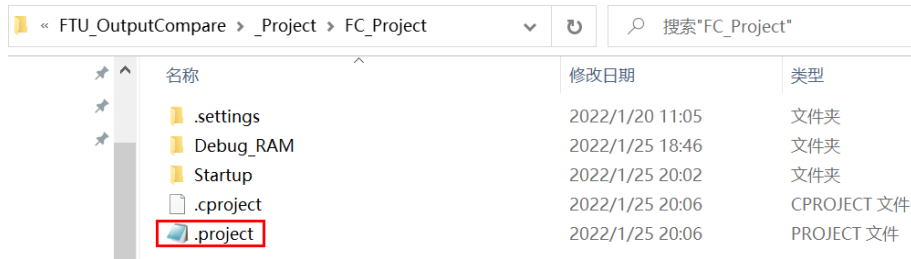
There are two methods to change the name of the project you imported:

Method 1 – Rename in the .project File:

- 1) Right-click the project name, and select **Show In > System Explorer** to open the directory where your project is located.



- 2) Open the .project file with an editor such as Notepad.

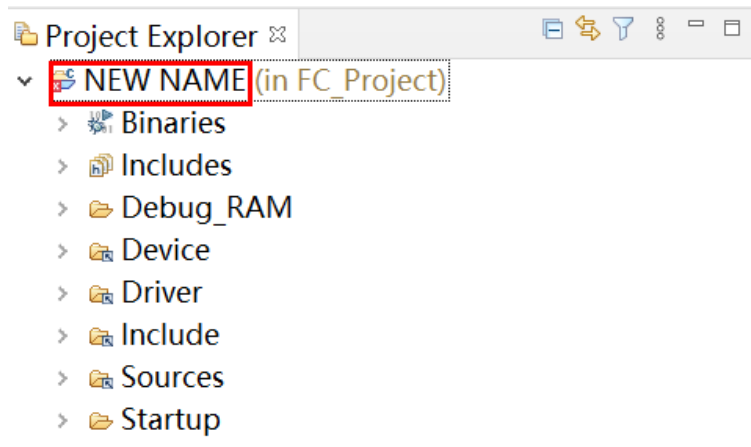


- 3) Change the project name enclosed within **<name>** and **</name>** to the one you want, and save it.

文件(F) 编辑(E) 格式(O) 查看(V) 帮助(H)

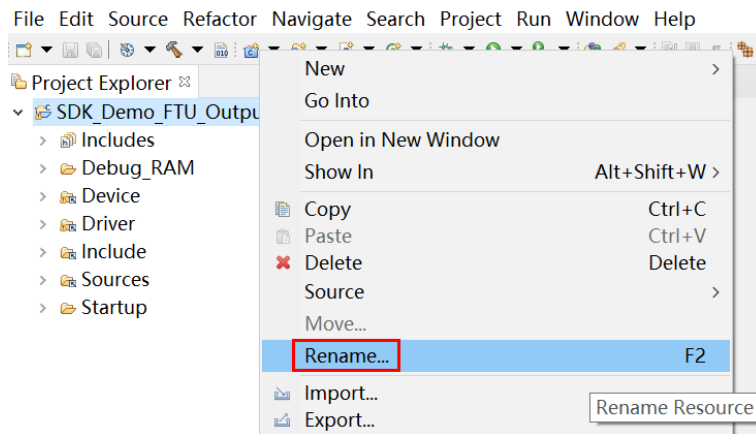
```
<?xml version="1.0" encoding="UTF-8"?>
<projectDescription>
  <name>SDK Demo FTU InputCapture</name>
  <comment></comment>
  <projects>
  </projects>
```

- 4) Rerun to the IDE window. Right-click the project name and select **Delete** from the pop-up menu to delete the project.
- 5) Import the project again by following the steps specified in Chapter 3.2 Import Project.
- 6) After the import is complete, the project name will be updated.

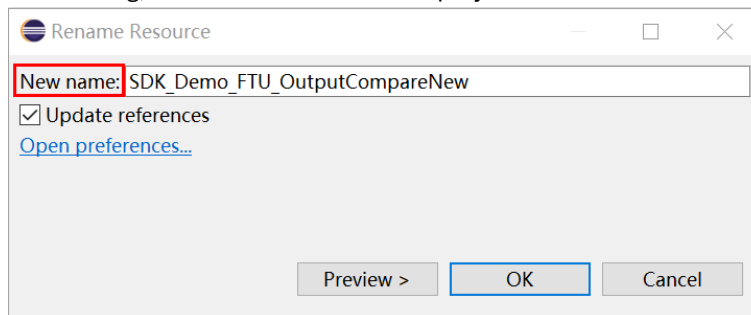


Method 2 – Right-clicking the Project:

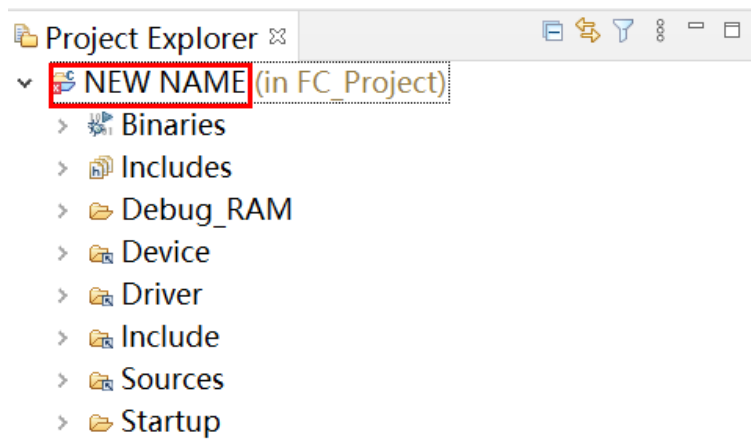
- 1) Right-click the project name, and select **Rename** to open the directory where your project is located.



- 2) In the **Rename Resource** dialog, enter a new name for the project into the **New name** text box.



- 3) Click **OK**, and the project name will be updated.



Note: It is **NOT** recommended to rename the project by using the Method 2, which will cause certain errors.

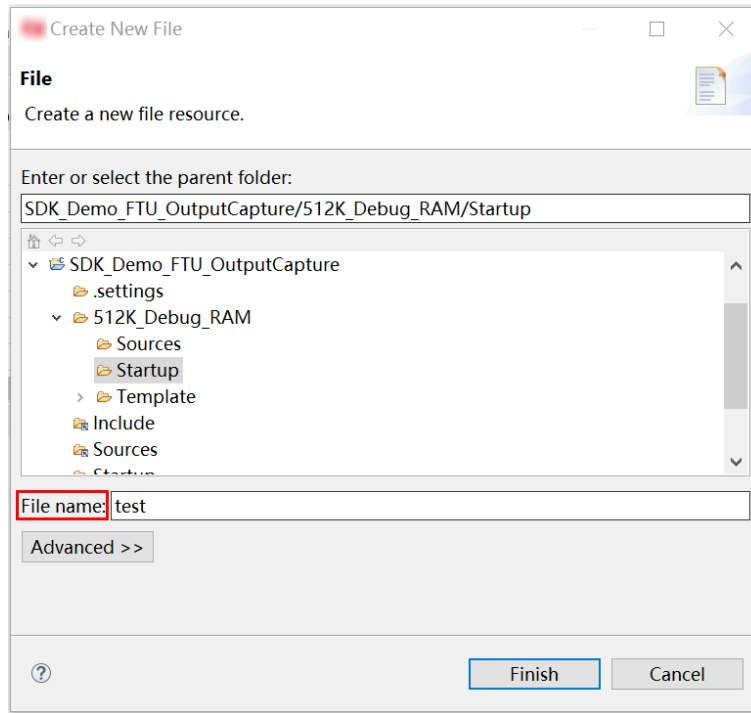
3.5.4 New File/Folder

There are two methods to create a file or folder in your project. Creating a file or a folder shares similar steps.

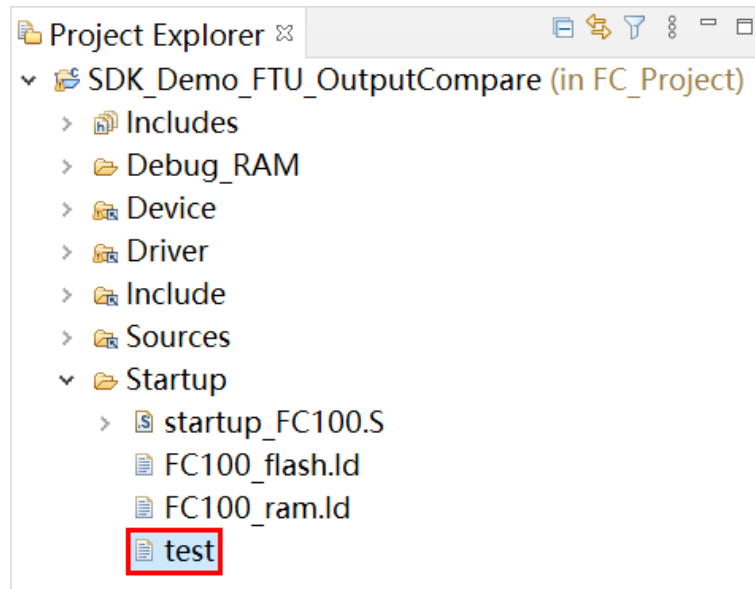
Method 1 – Right-clicking the Project:

Taking file creation as an example, the steps are as follows:

- 1) Right-click the folder where you want to create a new file.
- 2) Select **New > File**.
- 3) In the **Create New File** dialog, enter the file name you want to use into the **File name** text box.



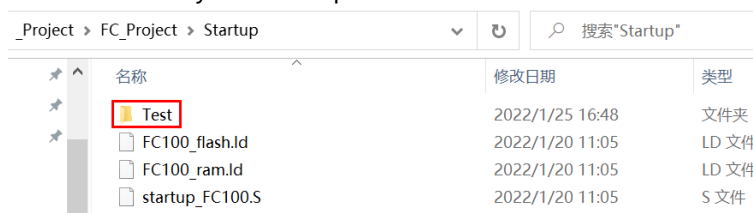
- 4) Click **Finish**, and the new file will be displayed under the folder you selected before.



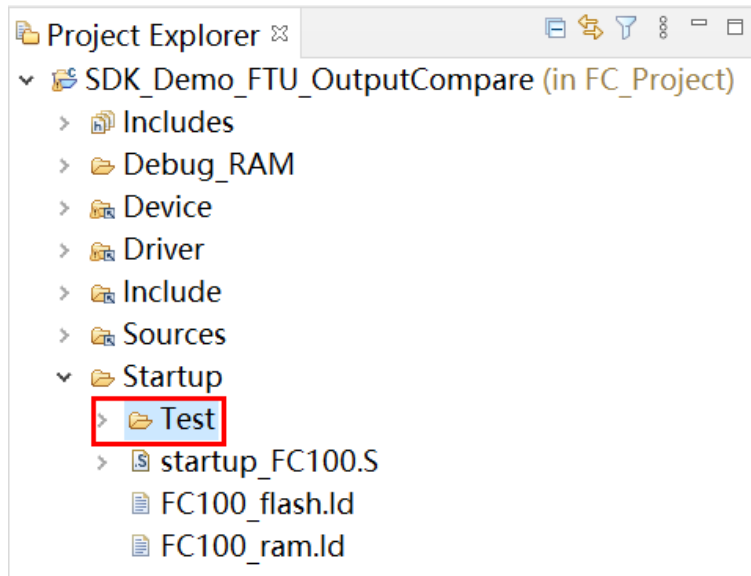
Method 2 – Creating a Local File/Folder:

Taking folder creation as an example, the steps are as follows:

- 1) Right-click the folder where you want to create a new folder, and select **Show In > System Explorer** to open the directory where the folder is located.
- 2) Create a folder in the location where you want to place it.



- 3) Right-click the project name and select **Refresh**, and the folder you created will be displayed.



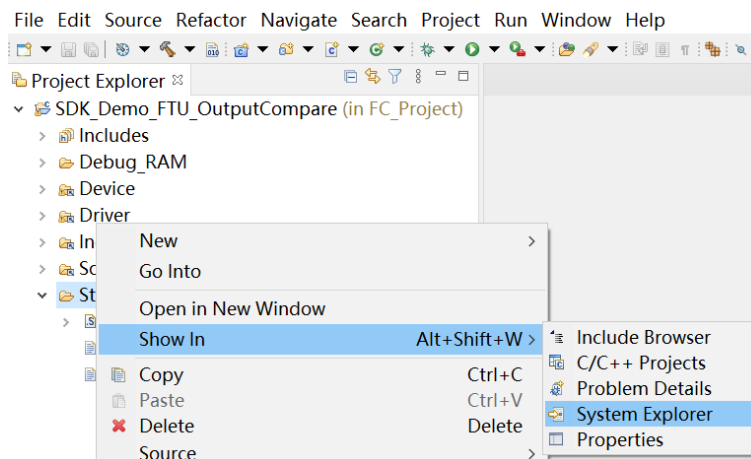
3.5.5 Add File/Folder

To add an existing file or folder to the project in the IDE, take the following steps:

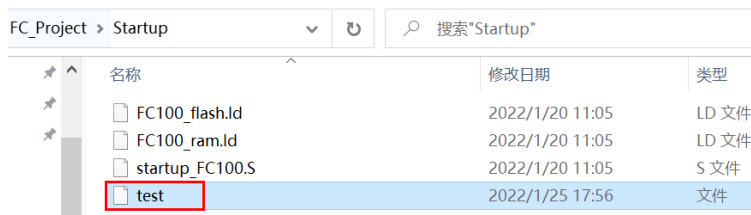
Method 1 – Copy to the Directory:

Taking adding a file as an example, the steps are as follows:

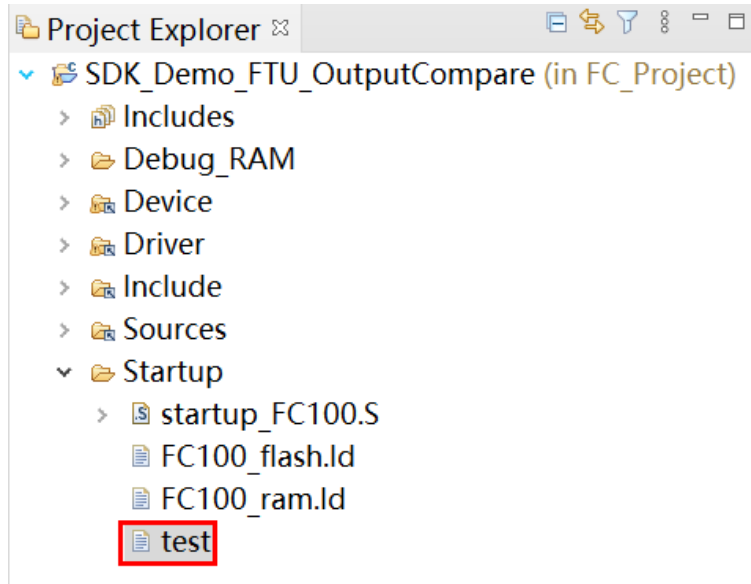
- 1) Right-click the folder where you want to create a new file, and select **Show In > System Explorer** to open the directory where the folder is located.



- 2) Copy the file you want to add to the location.



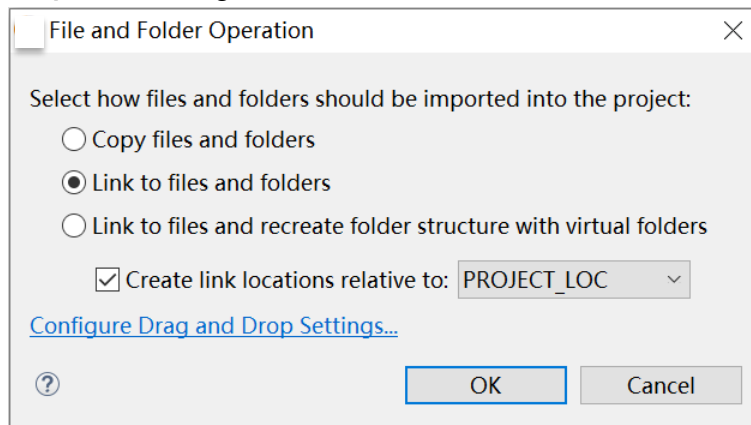
- 3) Right-click the project name and select **Refresh**, and the file you copied will be displayed.



Method 2 – Drag and Drop:

Take adding a folder as an example, the steps are as follows:

- 1) Drag an existing folder you want and drop it on the project folder to which you want to add.
- 2) In the **File and Folder Operation** dialog, select **Link files and folders**, and click **OK**.



- 3) The folder you dragged and dropped will be displayed on the left pane.

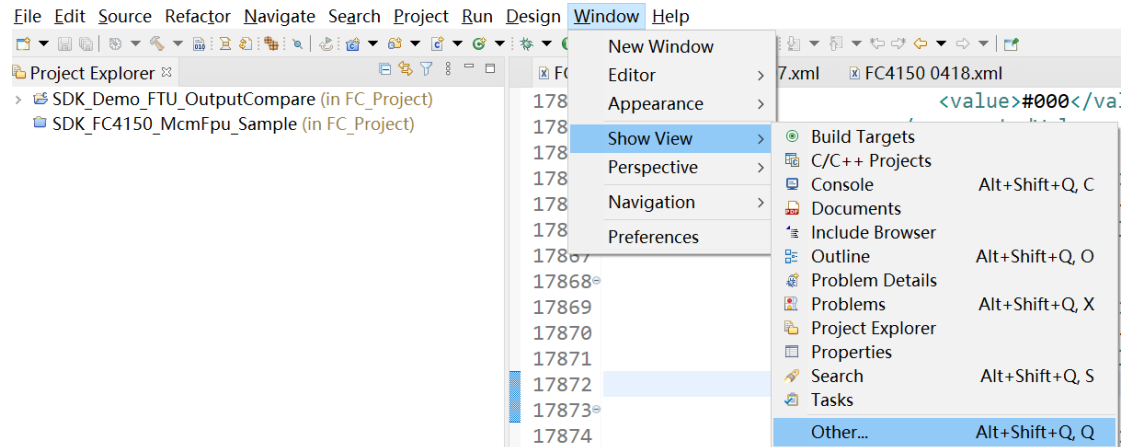
Note: If the folder you copied or created contains header files you may need, you can add them to the "Includes" folder. For details on how to add, refer to the Chapter 3.3 Add Header File to Includes.

Appendix A:

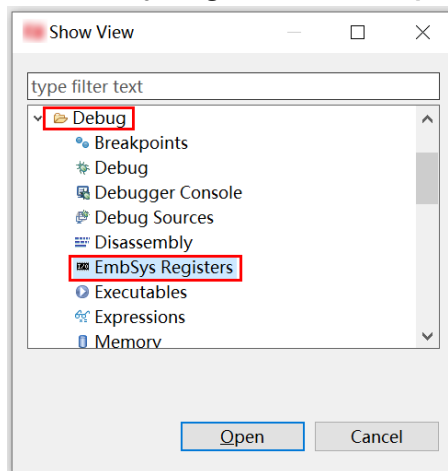
How to Load Register Definition File

To load the register definition file (in XML format) into the FC_IDE, take the following steps:

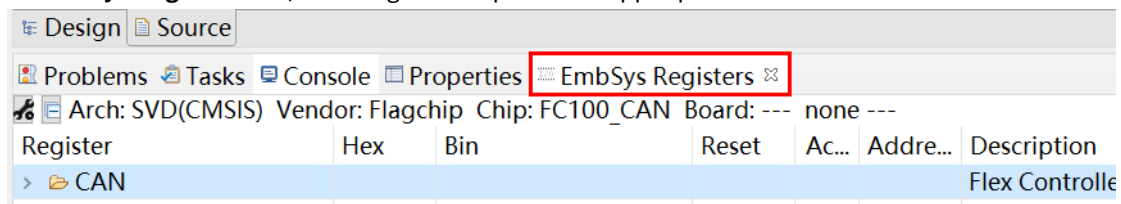
- 1) Select **Window > Show View > Other** on the menu.



- 2) In the pop-up **Show View** dialog, select **EmbSys Registers** and click **Open**.



- 3) The **EmbSys Registers** tab will be displayed at the bottom of the window. For a better view, you can click and press on the **EmbSys Registers** tab, and drag and drop it to the upper part of the window.



- 4) Click the  icon under the tab. In the pop-up **Preferences** dialog, select **Flagchip** from the **Vendor** dropdown list, and select **FC4150** from the **Chip** dropdown list.

